CELULAR AUTONOMA v2.1

The full organelle cast - a cell that can work, not just divide

7 optional, graftable smart-contract organelles,
wired through a gene-expression cycle.

68 tests passing - issues #19-#22 - PR #23 (stacked on #18)

What v2.0 still couldn't do

- v2.0 made the cell SELF-FUNDING (keeper credits) and HARVESTABLE (tithes, fees, sweeps).
- But it could only do two things: store capital and divide. It had no hands and no job:
  - nothing earned income from the world outside (no services to sell)
  - nothing grew the capital it held (no respiration of idle funds)
  - nothing kept house: no scheduling, no cleanup, no defense, no long-horizon storage
- Cell theory has an answer for every one of those gaps - an organelle.
- v2.1 installs the rest of the cast as OPTIONAL modules that plug into the living cell.

Cell theory -> contract theory

Mitochondrion	respiration -> ATP	deploys surplus capital to a yield venue, harvests accrued ATP back as food, breathes out when the cell needs liquidity
Ribosome	mRNA -> protein	translates approved 'genes' (templates) into fresh module clones - the in-cell evolution engine
Golgi	package & ship	scheduled, memo-tagged batch payments; ships a batch whole or not at all
Lysosome	digest & recycle	reclaims expired keeper credits into the cell, evicts condemned organs, sweeps dust
Peroxisome	detoxify	quorum-reported agents get flagged; their revenue is sequestered, not slashed
Vacuole	storage & turgor	timed tranches + jury-only quarantine custody + pressure() metric the harvest honors
Membrane	selective permeability	PRICED service receptors - the income organ: bots/agents/protocols pay per bind

Nucleolus = Colony template governance (shipped in v2.0) - cytoskeleton = the Nucleus registry (already anchors every organ)

Gene expression: how optional organs run themselves

1	keeper calls Chloroplast.express()	anyone can work the cell; any work pays
2	EnergyStore.startCycle()	cycle nonce++; the SPEND WINDOW opens
3	authorize(worker, budget)	budget = min(25% of remaining food, perWorkerCeiling), recomputed per organ
4	worker.act(budget) [try/catch]	organs do their jobs; a failing organ is skipped, never bricks the cell
5	endCycle() + keeper refund	window closes; unspent ceilings are now unreachable; trigger gets paid

Organelles NEVER hold treasury keys. Capability = a live registry fact: condemn an organ and its power dies instantly, no state to clean.

The capability model (what a grafted stranger can do)

Every money path is bounded 4x:

- 1. ROLE: spend() needs a CURRENT WORKER registration in THIS cell.
- 2. WINDOW: spend only works inside an active express() cycle - idle grafts are inert.
- 3. CEILING: per-cycle, nonce-scoped - unspent authorization simply dies.
- 4. RESERVE: discretionary balance only; keeper-owed ETH is untouchable by organs AND harvest.

The hole I found in my own draft

Ceilings alone leaked: a worker could call spend() between cycles against its leftover grant.

Fixed in review: startCycle opens the window, endCycle closes it. A malicious grafted organ can now extract AT MOST one per-worker ceiling per cycle - and only while the metabolism is deliberately activating it.

Test: 'the spend window is open only while the cell is working'.

The two income organs

Membrane - photosynthesis (outside)

- Cell advertises services: receptor = priced endpoint.
- AI agents / bots / protocols bind by paying the fee; funds flow straight to the cell's food.
- Per-payer ledger (what the Peroxisome meters), rate windows, overpay refunds, shield-checked admission.
- This is 'a useful function for the community' - the founding brief - as an on-market menu.

Mitochondrion - respiration (inside)

- Idle food earns nothing. Each cycle the mito deploys surplus to a yield venue (IYield adapter).
- Accrued interest = ATP: harvested home as FOOD, on the books via principal tracking.
- Breathes OUT (unstakes) when division or shipments need liquidity - float floor enforced.
- Real deployment = Aave v3 / ERC-4626 adapter behind the same 4-line interface.

The housekeeping organs

Golgi	schedule(to, amount, dueAt, memo) -> batch ships atomically when due & affordable. Bounded payroll for an agent economy. Known limit: a recipient that rejects ETH poisons its batch (cancel() is the lever; dead-letter queue is the fix).
Lysosome	keeper credits unpaid past the colony TTL are digested back into the cell (an un-pulled payment was never operationally owed); condemned organs get evicted from the registry; its own dust sweeps home.
Peroxisome	accusations need QUORUM of distinct reporters and enrich the accuser with NOTHING - the only consequence is isolation in the vacuole, released by a jury (the Colony). Reputation with skin in the game, not vigilante finance.
Vacuole	timed tranches flow back to the cell when mature; quarantine tranches never auto-release; pressure() is the turgor metric - the cell's long-horizon capital.

Exploratory results

68	+29	0
tests passing	v2.1 suite	regressions

Gas (scripts/organelle-study.js):

idle express() (no workers)	293,572 gas	~$0.0088
full 6-organ cast, cold cycle	852,663 gas	~$0.0256
full cast, warm cycle	606,712 gas	~$0.0182
mitosis WITH complete proteome (7 organs cloned+bound)	2,580,278 gas	~$0.0774

A daughter is born with its mother's whole proteome (6 workers + membrane, 10 registry entries, fresh single-hop clones). Running an entire multi-organ cell costs the price of a coffee crumb on Base/Sonic.

Honest notes from the review

- MockYieldVault is a TOY (linear bps, no share accounting, unguarded drain()) - never deploy it; real work = ERC-4626 / Aave adapter behind the same IYield surface.
- Peroxisome screening is heuristic reputation, not truth - consequence is isolation pending a COLONY JURY, never slashing, never accuser profit.
- lockQuarantine trusts any WORKER (registry has no per-name identity yet) - harmless: quarantine locks the depositer's own funds behind the jury.
- Golgi batches can be poisoned by one ETH-rejecting recipient (all-or-nothing per batch) - cancel() exists; dead-letter delivery is the next-iteration fix.
- digestExpiredCredits scans O(maxOps)/cycle - fine at colony scale, needs an index at internet scale.
- administer = owner config power; justified because the owner already has full capital power by design - its production replacement is a governance organelle, not a new key.

What a whole organism is now for

- Differentiation: same genome, different grafts - one colony now grows cell TYPES: relayer cells (membrane jobs), vault cells (mito posture), escrow cells (golgi schedules).
- The ribosome is the evolution engine: approved genes -> products -> graft; next step is genes chosen BY the colony's own governance organ.
- Epigenetics: an expression-policy module to decide which configured organs a daughter inherits.
- Production: Base Sepolia deployment, Aave IYield adapter, ERC-4337 paymaster so keepers need no wallet at all.
- The turntable run: a 30-cell colony with the full cast, harvesting to root daily, measured on its own earned income for N days.

Artifacts: issues #19-#22 - PR #23 (stacked on #18, OPEN for you) with 9-point inline review - 68 green tests - DESIGN.md organelle map - this deck (docs/).
Keet ping sent. Cell theory, compiled.